

**Ahsanullah University of Science and Technology****Course Title: Object Oriented Programming Lab****Course Number: CSE 1206****Online 7, Spring 2025 (Set B)****ID:** \_\_\_\_\_**Time: 30 Minutes****Marks: \_\_\_\_\_ / 10**

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| <p>ChatGPService is a software component that handles AI text generation.</p> <p>When generating text, it prints: <b>"Connecting to ChatGPT API to process Prompt: P"</b>, where P is a prompt (String).</p> <p>GeminiService is another software component that handles AI text generation.</p> <p>When generating text, it prints: <b>"Connecting to Gemini API to process Prompt: P"</b>, where P is a prompt (String).</p>                                                                                                                                      | 2 |
| <p>CodeEditor is a software that manages developer interactions and executes user prompts.</p> <p>When a prompt is responded to, it prints <b>"Connecting with AI models for prompt: P"</b>, where P is a prompt (String), and then <b>immediately triggers text generation for the AI service.</b></p>                                                                                                                                                                                                                                                             | 2 |
| <p><b>Design Constraint:</b> The system must be designed so that new AI services (such as ClaudeService or CopilotService) can be added in the future without modifying the CodeEditor class. To achieve this, CodeEditor must not instantiate ChatGPService or GeminiService directly, and it must not use if/else statements to choose between them. The specific AI service must be assigned when a CodeEditor object is created.</p>                                                                                                                            | 5 |
| <p>Implement this entire scenario using the necessary object-oriented principles (classes and interfaces). <b><u>Use only System.out.println for method implementations. Pass P as a method parameter (String).</u></b> Avoid code duplication.</p> <p>Demonstrate your design in the main method by:</p> <ol style="list-style-type: none"> <li>1. Creating a CodeEditor configured to use ChatGPService, and responding to a prompt.</li> <li>2. Creating a separate CodeEditor configured to use GeminiService, and responding to a different prompt.</li> </ol> | 1 |